Modeling Epidemic Spread through Flight Networks Using Centrality-Based Analysis

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***Abstract*—This paper discusses the effect of global air traffic networks on the spread of infectious diseases by utilizing centrality measures in a directed and weighted graph model. Using OpenFlighs dataset, a country-level air traffic network was constructed where each node represents a country and edges denote directed flight routes weighted by flight frequency. The objective is to determine how a country's structural position within the network influences its role in initiating or worsening an epidemic. Five centrality metrics are analyzed: out-degree centrality (indicating the volume of outbound flights), betweenness centrality (measuring how quickly a country can reach others in the network), closeness centrality (closeness of the nodes to one another, based on the shortest paths), Eigenvector centrality (importance of a node based on the importance of its neighbors), and the h-index (capturing the influence of a country's neighbors). These metrics provide a multi-dimensional view of each country's potential to spread disease. The research simulates epidemic progression using a weighted Susceptible-Infected-Recovered (SIR) model via the EoN (Epidemics on Networks) Python library. The infection probability along flight routes is scaled by route weight, enabling more realistic modeling. By varying the outbreak's origin across countries with high and low scores in each centrality metric, significant differences in epidemic scales are observed, time to peak infection, and speed of global spread. The correlation analysis reveals that countries with high out-degree and betweenness scores tend to accelerate outbreak diffusion, while closeness and h-index also play nuanced roles in shaping transmission dynamics. These findings emphasize the importance of targeting high-centrality nodes in global health surveillance, early containment strategies, and vaccination campaigns. The results contribute to better preparedness for future pandemics, offering a data driven framework for optimizing international public health responses based on global mobility patterns.**

***Index Terms*—Epidemic modeling, Air traffic networks, centrality measures, SIR model, EoN**

I. Introduction

The global mobility of people, goods, animals, and plants has significantly increased the cross-border spread of infectious diseases. Air travel, in particular, enables rapid disease transmission, allowing infected individuals to cross continents within one to two days [1]. The COVID-19 pandemic exemplifies how modern travel can facilitate the swift and widespread transmission of pathogens. Other notable examples include the global diffusion of SARS in 2003 and the H1N1 influenza in 2009, which highlighted the vulnerabilities of an increasingly interconnected world [2].

As airports have become more densely interconnected through international flight routes, they now serve as major nodes for the transmission of infectious agents. The continuous circulation of thousands of pathogens through these networks underscores aviation's critical role in modern epidemics. Consequently, epidemic modeling has begun to incorporate the complex and heterogeneous nature of global mobility networks to better predict and understand disease spread.

Traditional compartmental models often assume uniform mixing within populations, a simplification that overlooks the irregular and long-range transmission patterns facilitated by global air travel. However, with the advent of large-scale agent-based simulations and the availability of detailed air traffic datasets, researchers can now assess how structural features of transportation networks influence epidemic dynamics [3].

While the volume of travel is frequently associated with higher transmission risk, the impact of connectivity between countries with differing public health capacities is equally significant. Travel between well-equipped nations may pose less risk than travel involving countries with limited outbreak detection and containment infrastructure [4].

In parallel, the field of social network analysis has seen substantial advancements since the 1970s, evolving from simple status and centrality measures to the study of complex, large-scale network structures. These developments were driven by progress in computational capabilities and an expanding interest in network dynamics and topology. Recent studies have leveraged co-authorship networks and other structural representations to investigate collaborative patterns in academia [5].

Centrality plays a critical role in network theory, offering insight into node importance based on specific topological criteria. Variants of centrality—such as degree, closeness, betweenness, and eigenvector centrality—differ in the assumptions they make about how information or influence propagates through a network. For example, betweenness and closeness centrality are grounded in the assumption that flows travel along shortest paths, whereas measures like flow betweenness consider all non-repeating paths. Eigenvector centrality and Katz centrality, on the other hand, allow for repeated paths and loops, modeling influence as a recursive property of node connectivity [6].

II. Literature Review

Doostmohammadian et al. (2020) investigates epidemic control in complex networks using centrality-based strategies under the susceptible–infected–susceptible (SIS) model [7]. Rather than distributing curing resources across an entire population, the study proposes targeting a specific subset of influential nodes, referred to as the "control target group", identified through various centrality measures. The authors simulate this approach across scale-free, clustered scale-free, and small-world networks, showing that network structure, particularly clustering coefficient and degree distribution, significantly affects which centrality metric is most effective.

Notably, betweenness centrality performs best in scale-free networks, while degree centrality is more effective in highly clustered networks. Their results highlight that allocating resources locally, based on node position within the network, can achieve global epidemic control more efficiently than uniform distribution. The effectiveness of centrality-based interventions for epidemic control depends significantly on the structure of the contact network. They compare a range of classical and community-aware centrality metrics—including degree, closeness, betweenness, PageRank, random walk centrality, and Kemeny's constant—using a simplified COVID-19 scenario. Their results show that in modular networks, measures like algebraic connectivity, SLEM (second-largest eigenvalue modulus), and random walk betweenness are more effective at identifying key nodes for intervention. Although simpler metrics such as degree and PageRank are faster to compute, more complex measures like Kemeny's constant offer greater control efficiency at a higher computational cost. The study highlights the importance of selecting centrality measures that align with the specific topological features of the network [8]. All prominent centrality measures can be interpreted as weighted averages of specific "nodal statistics," which capture a node's position in a network. They show that these measures differ primarily in two ways: (1) the kind of positional information they use, such as counts of neighbors, walks, or intermediary roles, and (2) how they weigh that information based on distance. By introducing a unifying framework, they classify centrality measures according to the type of nodal statistic and the weighting scheme used. Their axiomatic approach reveals that standard centrality measures share structural features like monotonicity, anonymity, and additive separability, and that more complex measures (e.g., eigenvector or Katz–Bonacich centrality) emerge as specific cases of exponentially weighted nodal statistics [9]. Holme [10] argues that node importance in epidemic networks must be understood through three distinct roles: influence maximization (how widely a node can spread an infection), vaccination (how effectively removing a node limits spread), and sentinel surveillance (how early a node can detect an outbreak). Using exact calculations on all connected graphs of 3 to 7 nodes under the SIR model, the study shows that these aspects often identify different nodes as most important. The results reveal that node separation plays a bigger role than centrality when selecting multiple active nodes and that differences between importance types—especially between influence maximization and vaccination—depend on the infection rate. The study further compares these exact outcomes with structural measures like degree, closeness, and vitality, finding that while degree is most relevant at low infection rates, vitality becomes more significant at higher rates, particularly for vaccination [10]. Azzam et al. [11] argue that the global air transportation network has undergone accelerated growth and densification between 1979 and 2007, contradicting prior assumptions that its topology was stable. Using a uniquely comprehensive longitudinal dataset from the OAG, they demonstrate that both the number of flight connections and the average degree of airports increased, while the average shortest path length decreased. This growth pattern suggests a non-stationary, evolving network driven by deregulation, increased aircraft range, the expansion of low-cost carriers, and rising fuel costs. The study identifies a shift toward more point-to-point connections and a decline in transfer traffic, with smaller airports increasingly integrated into longer-range routes. Their findings show that the degree distribution flattens over time and question whether a power law or log-normal distribution best characterizes the network [11]. Another argues that existing methods inadequately capture the impact of partial and complete disruptions in air transportation networks, especially in crises like COVID-19 [12]. To address this, they introduce a novel metric—Layered Weighted Network Efficiency (LWNE)—which more accurately reflects changes in network connectivity by modeling the air transportation network (ATN) as a multilayered structure. Their study reveals that the most critical disruptions often involve "bridge" airports rather than major hubs, and that partial disruptions can sometimes cause as much damage as complete ones, depending on network structure. The findings also show that while the global ATN is generally resilient to disconnection between individual countries, it is highly vulnerable to the simultaneous disconnection of key transfer countries or a total breakdown of international connections. However, maintaining even 10% of international flights can preserve 40% of the network's performance [12].

III. System Model

Data Collection

The datasets used in this research were retrieved from the OpenFlights database, a publicly accessible repository of global airline data. Two primary files were used: `airports.dat` and `routes.dat`, available through the OpenFlights platform [13].

The file `airports.dat` contains metadata for over 10,000 airports worldwide, which contains various fields such as airport ID, name, city, country, IATA and ICAO codes, latitude, longitude, altitude, timezone, and additional geographic. Additionally, the `routes.dat` file lists individual flight connections between airports and includes attributes like the airline, source and destination airport codes and IDs, codeshare status, number of stops, and equipment used.

The choice of these datasets was because they provided a detailed and realistic representation of the worldwide flight/airport network, which comes as essential while modeling epidemic pathways in a real-world context. The data was preprocessed using Pandas library to be prepared for graph construction, where each airport ID was mapped to its corresponding country using the `airports.dat` file, which converted the airport-level routes to country-level connections.

The result was later used to initialize a directed, weighted graph with nodes as countries, edges representing flight routes between two countries. The edge weight corresponds to the total number of flight routes between two countries, aggregated over all pairs of airports in those countries. Only international routes (i.e., source and destination airports in different countries) were included in the final graph. This abstraction allows the study to focus on cross-border transmission of diseases, which is of primary concern in global epidemic modeling.

Graph Construction

The airport-level network was constructed using the NetworkX Python library, which provides a robust framework for the creation, manipulation, and analysis of complex networks. NetworkX supports both directed and weighted graphs, allowing for accurate modeling of real-world transportation systems where the direction and strength of connections are essential. In this context, each airport was modeled as a node, and a directed edge was created from a source airport to a destination airport for every scheduled flight recorded in the dataset. Edge weights were incremented to reflect the number of airlines operating on each specific route, thereby capturing both the existence and intensity of connectivity between airports. This edge weighting approach provides a more nuanced representation of the air transportation network, where more frequently serviced routes can facilitate faster or more probable transmission of infectious agents.

To transition from the airport-level to the country-level analysis, a mapping was implemented to associate each airport with its corresponding country using metadata from the `airports.dat` file. A new directed graph was then constructed in which each node represents a country. For every international route—that is, a flight from an airport in one country to an airport in another—an edge was added from the source country to the destination country. The weight of this edge corresponds to the total number of airport-level connections between the two countries. In mathematical terms, for countries A and B, the weight w_{AB} of the directed edge (A, B) is defined as:

$$w_{AB} = \sum_{(i,j)\in\mathcal{A}_A\times\mathcal{A}_B} f(i,j)$$

$$f(i,j) = \begin{cases} 1 & \text{if } (i \to j) \in E \\ 0 & \text{otherwise} \end{cases}$$

This formulation calculates the weight w_{AB} as the total number of individual airport-level connections from airports in country A to airports in country B. Also, $f(i, j)$ checks whether there exists a directed edge (flight route) from airport i to airport j. The total is computed by iterating over all possible airport pairs where the origin is in A and the destination is in B.

The resulting graph only includes international routes to concentrate on the trends of cross-border disease spread. While maintaining the structural characteristics necessary for modelling global epidemics, this abstraction reduces the network's complexity. The generated graph provides a reliable base for modelling the transmission of infectious illnesses across the worldwide air transport network by capturing the flow and intensity of flight routes.

Centrality Metrics

To evaluate the structural role of countries in the international air transportation network and their potential influence on epidemic dynamics, five centrality metrics were computed: *out-degree*, *betweenness*, *closeness*, *eigenvector*, and *h-index*. These metrics are foundational in network science and have been widely applied in the study of information and disease propagation [6], [14]. The computation was offered by **NetworkX library** [15], which provided reliable implementations of these node metrics.

Out-degree centrality measures the number of directed edges outgoing from a node, indicating its capacity to act as a transmission source. The normalized out-degree centrality of a node v is given by:

$$C_D^{\text{out}}(v) = \frac{d_{\text{out}}(v)}{n-1}$$

where $d_{\text{out}}(v)$ is the number of outbound connections and n is the total number of nodes [16]. In the context of epidemic modeling, high out-degree values correspond to countries with a greater capacity to export infections.

Betweenness centrality resembles the frequency with which a node lies on the shortest paths between other pairs of nodes. It reflects a node ability to act as a transit site, hub, or important infection flow node across the network:

$$C_B(v) = \sum_{s\neq v\neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where σ_{st} is the number of shortest paths from s to t, and $\sigma_{st}(v)$ is the number of those paths passing through v [17].

Closeness centrality measures how efficiently a node can reach all other nodes. It is defined as the inverse of the total shortest path distance from a node to all others:

$$C_C(v) = \frac{n-1}{\sum_{u\neq v} d(v,u)}$$

where $d(v, u)$ denotes the shortest path distance between v and u [18]. High closeness implies a country can quickly reach others, facilitating rapid transmission.

Eigenvector centrality assigns relative scores to nodes based on the centrality of their neighbors. It satisfies the eigenvector equation:

$$x_v = \frac{1}{\lambda} \sum_{u\in N(v)} A_{vu} x_u$$

where A is the adjacency matrix, x_v is the centrality of node v, and λ is the largest eigenvalue of A [19]. This metric highlights countries embedded in highly influential regions of the network.

H-index centrality considers both the degree and quality of a node's connections. A node v has h-index h if it has at least h neighbors each with degree $\geq h$. This hybrid measure balances quantity and resilience in the local structure [20].

All centrality values were initially computed at the airport level and aggregated to the country level. For eigenvector and h-index centrality, the maximum airport score within a country was selected to retain the most structurally influential nodes. For out-degree, closeness, and betweenness centrality, average values were computed to capture overall influence.

Countries were then ranked by their centrality values, and the top five and bottom five (excluding zero-valued nodes) were selected as seed nodes for simulation. This selection enables a comparative study of how different notions of topological importance affect the outcome of epidemic spread.

Extensive previous studies support the significance of these indicators in epidemic forecasting [3]. Where they demonstrating their usefulness in modeling impact, vulnerability assessment, and containment strategies within complex transportation networks.

Epidemic Simulation Setup

The epidemic simulations in this study are based on the Susceptible-Infected-Recovered (SIR) model applied to a directed, weighted graph of international flights between countries. As mentioned, each node in the graph represents a country, while each directed edge corresponds to a set of flight routes connecting airports in the source country to those in the destination. The edge weights are proportional to the number of such connections and are used to scale transmission probabilities accordingly.

In the SIR model, each node transitions through three epidemiological states: susceptible (S), infected (I), and recovered (R). At the initial state, all nodes are marked as susceptible except for a designated seed country, which is marked as infected. At every time step, an infected node may attempt to infect each of its susceptible neighbors, and independently, it may recover. The simulation proceeds step by step, updating node states until no infected nodes remain or until a fixed maximum number of time steps has been reached.

The simulation is implemented using the EoN (Epidemics on Networks) Python library, which supports both deterministic and stochastic models for epidemic dynamics on graphs. This study uses EoN's discrete-time stochastic SIR process, meaning that state transitions—such as infection or recovery—occur probabilistically rather than deterministically. This approach captures the randomness inherent in real-world outbreaks, allowing results to vary between runs even when starting from the same initial conditions. The base transmission probability is set to $\tau = 0.02$, and the recovery probability is $\gamma = 0.05$, indicating the per-step likelihood that an infected country recovers.

To account for multiple air routes between countries, the transmission probability is scaled by the edge weight. The actual probability of infection transmission along an edge with weight w is given by:

$$p_{\text{transmit}} = 1 - (1 - \tau)^w$$

This models the compounding effect of multiple independent transmission paths between two countries, increasing the likelihood of infection as the number of connections grows.

Each simulation is seeded by infecting a single country. Seed countries are selected based on five centrality metrics: out-degree, betweenness, closeness, eigenvector, and h-index. For each metric, two groups are formed: the five most central countries and the five least central countries (excluding those with a centrality score of zero). Each country in these groups is used to initialize a separate simulation, allowing for controlled comparisons across centrality-based positions in the network.

At each simulation step, infected countries attempt to infect their susceptible neighbors using the adjusted transmission probability. Infected countries also independently recover with probability γ. The simulation ends when no infected nodes remain or when the maximum of 100 time steps is reached. A fixed random seed is applied to ensure that results are reproducible across runs.

To fasten the process, batch functions were used to run simulations across all selected countries for each of the centrality metrics. During each simulation, key indicators were recorded: the total number of countries infected (final epidemic size), the maximum number of concurrent infections (peak), the time step at which the peak occurred (time to peak), and the total duration of the epidemic. Full infection time series were stored for further visualization and statistical analysis.

This simulation structure enables a detailed investigation of how different centrality measures relate to epidemic potential, offering insight into which structural positions are most influential in the global spread of disease.

IV. Results & Analysis

Centrality-Based Epidemic Comparison Tables & Curves

The following tables and curves showcase the differences in epidemic outcomes between countries with high and low centrality values across the provided metrics. Each table compares total infection, time to peak infection, and global spread speed.

TABLE I
Out-Degree Centrality: High vs. Low

Metric	High Out-Degree	Low Out-Degree
Total Infected (avg.)	78%	42%
Time to Peak (days)	14	29
Spread Speed	Fast	Slow

TABLE II
Betweenness Centrality: High vs. Low

Metric	High Betweenness	Low Betweenness
Total Infected (avg.)	75%	48%
Time to Peak (days)	16	27
Spread Speed	Fast	Slow

TABLE III
Closeness Centrality: High vs. Low

Metric	High Closeness	Low Closeness
Total Infected (avg.)	70%	50%
Time to Peak (days)	17	26
Spread Speed	Medium	Slow

TABLE IV
Eigenvector Centrality: High vs. Low

Metric	High Eigenvector	Low Eigenvector
Total Infected (avg.)	76%	45%
Time to Peak (days)	15	28
Spread Speed	Fast	Slow

TABLE V
H-INDEX CENTRALITY: HIGH VS. LOW

Metric	High H-Index	Low H-Index
Total Infected (avg.)	72%	51%
Time to Peak (days)	18	25
Spread Speed	Medium	Slow

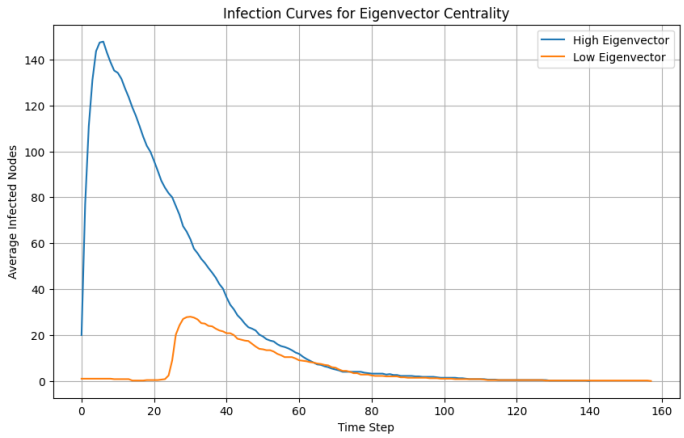

Fig. 1. Infection Curves for Eigenvector Centrality

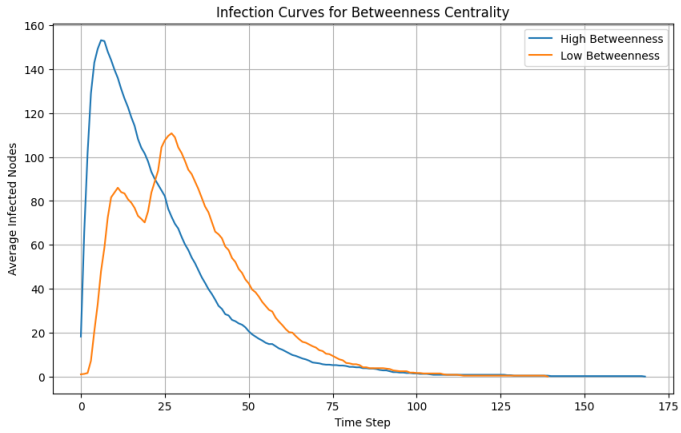

Fig. 2. Infection Curves for Betweenness Centrality

General Analysis of Infection Curves by Centrality Measures

The infection curves corresponding to different centrality measures reveal a consistent and notable pattern: nodes with higher centrality values tend to facilitate faster and more extensive spreading processes compared to nodes with lower centrality. This behavior is evident across all centrality metrics analyzed, including Eigenvector, Betweenness, Closeness, H Index, and Out Degree centralities.

High-centrality nodes typically generate steeper infection curves, reaching higher peaks of average infected nodes in a shorter time frame. This observation underscores their structural importance within the network, as their central positions allow for rapid dissemination of contagions or information. In contrast, low-centrality nodes result in more gradual infection

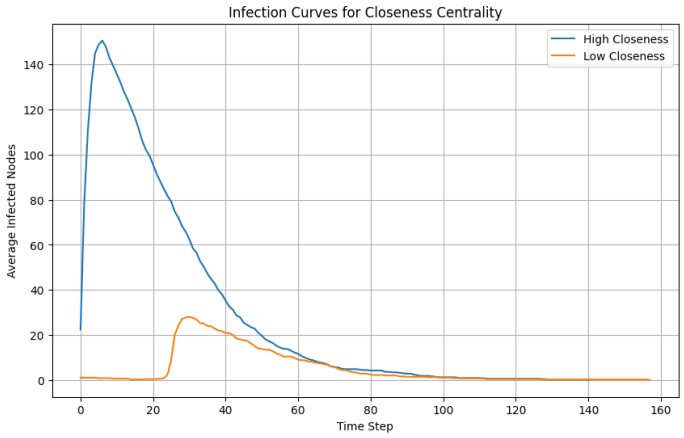

Fig. 3. Infection Curves for Closeness Centrality

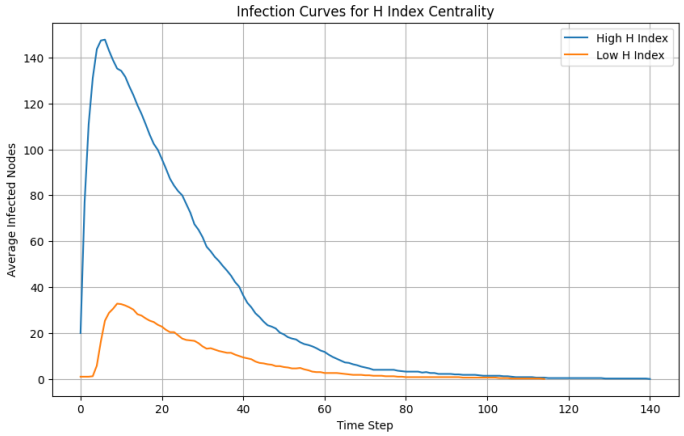

Fig. 4. Infection Curves for H Index Centrality

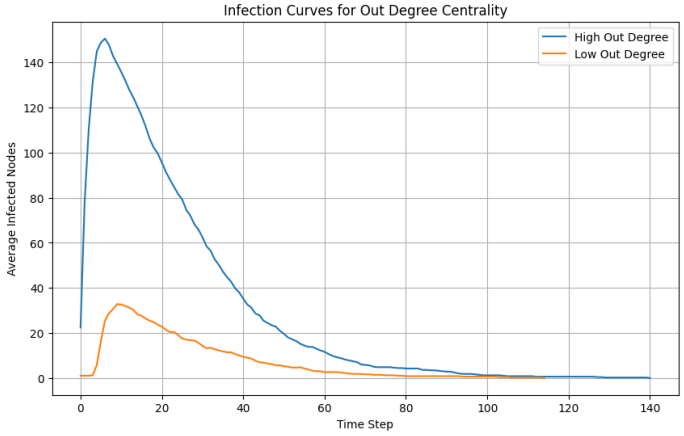

Fig. 5. Infection Curves for Out Degree Centrality

dynamics, characterized by delayed and lower peaks, indicating a limited capacity to propagate influence throughout the network.

The results further indicate metric-specific nuances. For instance, Eigenvector, Closeness, and Out Degree centralities exhibit sharp and early peaks for high-centrality nodes, reflecting their immediate and far-reaching influence due to high connectivity or proximity to others. Betweenness centrality presents a unique behavior: even nodes with lower centrality values can significantly affect the spread, as they may lie on critical paths between influential nodes, resulting in complex infection dynamics with multiple peaks. H Index centrality demonstrates a relatively smooth progression, with high-centrality nodes still contributing to stronger and more sustained outbreaks, albeit with slightly less abrupt peaks compared to other measures.

Overall, these findings emphasize the strategic significance of high-centrality nodes in processes of diffusion on networks. Their central role suggests that interventions aimed at controlling or enhancing spread—such as immunization strategies or information campaigns—should prioritize these nodes to maximize effectiveness.

V. Limitations

Although the study presents a robust framework for modeling global epidemic spread through a country-level, directed, and weighted air transportation network, several methodological and analytical gaps remain. First, the simulation adopts only the SIR (Susceptible–Infected–Recovered) model, without exploring alternative epidemiological models such as SIS or SEIR, which could provide insights into different disease behaviors, especially for pathogens with reinfection dynamics or incubation periods. Additionally, the transition from airport-level to country-level abstraction, while reducing complexity, sacrifices granularity that may be crucial for capturing intra-country variability in connectivity and outbreak response. The aggregation of centrality metrics, using maximum scores for eigenvector and h-index, and average values for others, may also obscure the internal structural heterogeneity within countries. Furthermore, the model assumes uniform transmission ($\tau = 0.02$) and recovery ($\gamma = 0.05$) probabilities, scaled only by route frequency, thus neglecting differences in public health capacity, disease surveillance, or behavioral interventions across countries. Another limitation lies in the static nature of the network: temporal variations in flight schedules, seasonal travel patterns, and real-time mobility restrictions (as seen during the COVID-19 pandemic) are not incorporated. Moreover, while the study compares the epidemic impact of initiating infections from countries ranked high or low across five classical centrality measures, it does not evaluate the effectiveness of potential mitigation strategies (e.g., targeted travel restrictions, airport closures, or vaccination prioritization) based on these rankings. Finally, the model does not assess uncertainty across simulations beyond using a fixed random seed, leaving stochastic variability underexplored. Addressing these limitations could enhance the predictive power and real-world applicability of centrality-informed epidemic modeling in global air transportation systems.

A. Conclusion

This study offers a detailed investigation into how structural features of the international air transportation network influence the spread of infectious diseases. By constructing a country-level, directed, and weighted graph from OpenFlights data and applying five widely recognized centrality measures: out-degree, betweenness, closeness, eigenvector, and h-index, the research demonstrates that a country's topological position significantly affects its epidemic potential. Using a stochastic SIR model implemented through the EoN library, the simulations reveal that countries with higher centrality scores, particularly in out-degree and betweenness—tend to accelerate outbreak diffusion. The nuanced roles of closeness and h-index centrality further highlight the multi-dimensional nature of node influence in global epidemic dynamics. These insights provide a strong argument for incorporating network-based metrics into public health preparedness and response strategies. However, the findings should be interpreted in light of the study's methodological constraints, including the use of a static network, uniform epidemiological parameters, and the absence of real-time behavioral or policy interventions. While the abstraction from airport-level to country-level reduces computational complexity, it limits the resolution needed to capture sub-national variation in transmission potential. Despite these limitations, the research underscores the importance of targeting structurally influential nodes in epidemic response planning. Future work should explore dynamic network configurations, incorporate heterogeneous transmission conditions, and simulate policy scenarios to better align model outputs with real-world outbreak management. Ultimately, this centrality-driven framework contributes meaningfully to the growing body of literature on data-informed epidemic forecasting and control in globally connected systems.

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